Theory-laden Evidence and Incommensurability in a Bayesian Setting

Elijah Spiegel

1 Ornithological Example

Theory-laden evidence can lead to incommensurability of scientific theories even in tame cases. Suppose two ornithologists have a disagreement over whose preferred theory is correct. They decide to settle their differences by conducting an observational study in Western Mauritania. They make observations separately, so as to cover more ground, recording data such as species, weight, wingspan, etc. When they return they realize, to their dismay, that one ornithologist classified using a 2023 guidebook and the other using a 2025 guidebook. (In retrospect, they realize that this was because their respective theories are committed to the respective classification schemes.) This turns out to be a problem.

Each year, ornithological societies release updated species classifications. These regularly result in 'lumps' and 'splits'. A lump occurs when two or more species of bird previously considered distinct are lumped into one species. A split occurs when two or more groups considered one species, typically subspecies, are split into different species. A 2023 guidebook distinguishes the African reed warbler from the Eurasian reed warbler, which a 2025 guidebook lumps together as the Common reed warbler (*Acrocephalus scirpaceus*), and a 2023 guidebook groups together the Atlas wheatear (*Oenanthe seebohmi*) and the Northern wheatear (*O. oenanthe*), which are distinguished by a 2025 guidebook.[1] These are all birds that can be found in Western Mauritania, and both ornithologists ended up recording data on them according to their respective classification schemes.

The theories the ornithologists are arguing about are, for theoretical reasons, committed to different respective species classifications. Theory

[1] See figure 1 on page 9 for a visual aid.

A takes the 2023 species concepts to be correct and Theory B takes the 2025 species concepts to be correct. Theory A does not make the Northern wheatear–Atlas wheatear distinction, and Theory B does not make the African reed warbler–Eurasian reed warbler distinction. This can lead to problems when comparing the theories. Each theory, we can suppose, does well at accounting for patterns it sees in the data that the other theory does not even recognize due to not distinguishing individuals in the same way. From each theory's perspective, it is doing better at explaining the evidence while the other is claiming an unearned advantage by accounting for something either irrelevant or meaningless.

This story gives us a toy example of theory-ladenness of evidence and incommensurability of theories, but a realistic one. Scientists in the real world face problems synthesizing bodies of evidence shaped by different theoretical backgrounds, and these differences can lead to difficulties in comparing theories. This is a problem worth studying in detail.

2 Theories and Theory-laden Evidence

To say that evidence is theory-laden means that one's theoretical background partially determines one's evidence.[2] In Kuhn's nicer prose, "the proponents of competing paradigms practice their trades in different worlds. [...] Practicing in different worlds, the two groups of scientists see different things when they look from the same point in the same direction." (Kuhn et al. 1970) How this is understood can vary from triviality to hyperbole. Read weakly, this could just mean that different researchers prioritize different facts. One cancer researcher may 'practice in a world' of genetics and transcriptomes while another may 'practice in a world' of biomechanics and tissue architecture. Emphasizing different facts may lead them to support different conclusions, even when looking at the same overall body of evidence. Read very strongly, Kuhn might mean that what facts are true in our world partially depends on what our theoretical commitments are.

Theory-ladenness of evidence generates two problems for the objectivity of science. The first problem is a commensurability problem. If evidence is theory-laden then there is no theory-neutral basis on which to compare theories. Different theories may, from the same experiential basis, generate

[2]I am consciously departing from the standard phrase "theory-ladenness of observation" since I want to treat theory-ladenness from non-perceptual sources as well.

differently theory-laden bodies of evidence that disagree about how theories compare. The second problem is a convergence problem and is really a corollary of the first problem. If evidence is theory-laden then there is no guarantee that inquirers who start with different theories, and so follow different evidential cues, will converge to the same picture of the world (if they even converge at all). There is *a fortiori* no guarantee that they will both converge to a true picture of the world assuming that this entails converging to the same picture.[3] This problem is a special consequence of the first problem that shows how high the stakes are.

Theory-ladenness of evidence has many potential sources. To name just a few examples in roughly descending order of contentiousness, different theories may yield differently theory-laden bodies of evidence because they: disagree about which evidential propositions are meaningful, disagree about what classification scheme to use, disagree about whether some evidential propositions matter enough to be included in the evidence, disagree in how (or whether) they theorize about the reliability of a measuring device, disagree about which phenomena are more important to explain, influence practitioners through expectation bias, assign different meaning to the same word, or disagree about which experiments are worth doing.[4] The contentiousness of the source is roughly proportional to how radical it makes the consequences of theory-laden evidence. The last three examples in the list, in particular, lead to problems in practice rather than in principle for theory development and comparison.[5] I will set aside these less extreme sources of theory-ladenness and focus instead on sources that are supposed to raise more fundamental difficulties.

With so many different sources of theory-ladenness, how should we develop a maximally general explication of theory-laden evidence? I suggest looking for inspiration in recent work in the Bayesian literature on awareness growth. So, first, we must rehearse the standard Bayesian machinery.

The fundamental object of probability theory is a probability space $\langle \Omega, \mathcal{F}, P \rangle$ consisting of (all to be defined shortly) a set Ω, a sigma algebra $\mathcal{F}$ on Ω, and a probability measure P on $\langle \Omega, \mathcal{F} \rangle$. For Ω, the only relevant

[3]This problem might not arise for a very strong reading on which the different theories make different worlds.

[4]See Bokulich (2020) for an extended discussion of how theory-ladenness operates in data-intensive sciences through sources not listed here.

[5]Franklin (2015), for example, argues that all problems of theory-ladenness are practical, not philosophical, problems solved on a case-by-case basis by working scientists.

mathematical property is that it is a set. A sigma algebra $\mathcal{F}$ on Ω is a set of subsets of Ω containing Ω and closed under countable union and complement relative to Ω. A probability measure P on $\langle \Omega, \mathfrak{F} \rangle$ is a countably additive function from $\mathcal{F}$ into $[0, 1]$ such that $P(\Omega) = 1$.

Probability spaces admit many philosophical uses. In Bayesian epistemology, we usually use probability spaces to represent the degrees of belief of rational agents. However, in this paper and following the example of, for example, Romeijn (2013) and Henderson et al. (2010), I will use probability spaces to represent scientific theories. The proposed explication of scientific theories can be thought of as a probabilistic version of the Received View of scientific theories as axiomatized theories in a logical language.[6] (Suppe 1977) The role of the sentences is now played by the propositions in $\mathcal{F}$ and, instead of having sentences that are provable, refutable, or independent of the axiomatization, we now have propositions that are assigned probability one, zero, or a non-extreme value by P.

Using probability spaces to represent scientific theories has two nice benefits. First, it makes it clear where the probability assignment comes from, thus sidestepping the Problem of the Priors.[7] Second, although this formalization like the Received View is an idealization, unlike the Received View the actual practices of scientists can be seen as working towards this ideal. A survey of publications in almost any scientific discipline suggests that the arc of science bends towards more, not less, statistics, and the more that scientists extend and develop their theories to make them more statistically sophisticated and admit more statistical testing the more their theories are brought closer to this ideal.[8] Instead of trying to impose a particular formal straitjacket on science, we analyze scientific practice with a formal ideal it sets for itself.

Let me now say a little more about the philosophical interpretation of the components of a probability space. As usual, we can think of Ω as the set of

[6]As an aside, I do not claim the Received View was actually anyone's received view except perhaps Suppe's.)

[7]The Problem of the Priors rears its head again if we try to model an agent who must adopt a prior over theories, but for reasons that will become apparent an even more fundamental problem for this modeling task is building an algebra in which all of the theories can live.

[8]This view might require the assumption that all (good) statistical testing is or approximates Bayesian statistics, but I suspect a more frequentist-tolerant story can be made to work. Developing this is beyond the scope of this paper.

possible worlds. I think of possible worlds from the perspective of a modeler rather than metaphysics, but nothing hinges on this. I assume that all of the theories we will discuss are, as probability spaces, defined on some common Ω.

The interpretation of the sigma algebra $\mathcal{F}$ is especially significant in this paper. Easwaran and Nielsen (2025), in the context of modeling an agent, describe the elements of $\mathcal{F}$ as "the propositions that the agent grasps."[9] (Easwaran and Nielsen 2025, p. 1096) Along similar lines, we can understand the elements of $\mathcal{F}$ as the propositions that the scientific theory 'grasps'. This includes, for example, all of the propositions that the theory makes predictions about, propositions about theoretical entities discussed by the theory, and propositions that express evidence relevant to the theory.

Finally, the probability measure P is that of a scientific theory fully articulated to assign probabilities to the propositions in $\mathcal{F}$. Of course, even a fully mature scientific theory may not uniquely determine all these probabilities. For example, the scientific theory may be deterministic but not specify extreme values for some propositions. In that case, we should think of P as specifying the probabilities according to the scientific theory along with a reasonable prior over what the theory does not specify, but only the minimal amount necessary to make the theory yield predictions. Although more arbitrary than some philosophers would like, this is in line with the actual practices of scientists who adopt priors over parameters to make Bayesian inference possible.

Now, how should we explicate theory-laden evidence? Since differently theory-laden bodies of evidence arise from (of course) different theories, it is natural to take a cue from recent Bayesian work on awareness growth studying how to model agents' becoming aware of different theoretical possibilities. While details vary, the central insight is to represent differences in conceptual-theoretical resources as differences in the algebras over which probability measures are defined. (De Canson 2024) (Steele and Stefánsson 2021) (Wenmackers and Romeijn 2016)

To convince ourselves that the techniques from the awareness growth literature are promising for explicating theory-ladenness, we can consider some examples that show how the differences in conceptual resources that produce differently theory-laden evidence are the same kinds of differences

[9]More specifically, they say that it is the set of propositions the agent at least implicitly grasps, which may be generated by a base of propositions that the agent directly grasps.

handled by the awareness growth approach. Following a taxonomy from de Canson, we can distinguish three kinds of awareness growth. In orthogonal awareness growth, an agent becomes aware of a new proposition that is logically independent of previously grasped propositions. So, for example, in the context of describing bacteria a microbiologist might come to grasp the category of gram positive as a relevant category that is neither entailed nor precluded by any categories she already grasps. This is represented by refinement of the agent's sigma algebra with a proposition that is compatible with all (non-empty) propositions already in the sigma algebra. In internal awareness growth, an agent becomes aware of a new proposition that entails some proposition she already grasps. So, for example, an ornithologist might come to distinguish two species of bird based on plumage that she previously considered to be the same. This is represented by a similar refinement as in orthogonal awareness growth, but with a new proposition that does not 'cut across' all existing non-empty propositions. Finally, in lateral awareness growth, an agent becomes aware of a new proposition that is inconsistent with the previously grasped propositions. So, for example, systematic errors may lead a scientist to seriously entertain the possibility that her electron microscope is on the fritz. There is disagreement over how to represent this kind of awareness growth, but I would treat it as sigma algebra refinement that results in differentiation within a set of possible worlds called the 'catch-all' that is not differentiated by any existing propositions and has a privileged interpretation as the occurrence of some unforeseen possibility (and is typically assigned probability zero).[10] So lateral awareness growth can also be understood as a refinement of the sigma algebra. As the given examples are intended to illustrate, each kind of awareness growth may lead to the kind of change in conceptual resources that can produce differently theory-laden evidence given the same epistemic affordances by the world. So, the awareness growth literature's insight of modeling conceptual resources with sigma algebras seems a good starting point for an explication of theory-laden evidence.

The awareness growth literature does not discuss theory-laden evidence. However, we can take the approach of modeling conceptual resources as

[10]The details here are not important for understanding the paper, but to be clear I would think of lateral awareness growth not by adding worlds to Ω but by adding graspable distinctions among worlds that are distinguished from a modeler's perspective even if they are not distinguished from the agent's perspective prior to awareness growth.

sigma algebras as a starting point to explicate theory-ladenness.[11] As de Canson puts it while describing refinement of the sigma algebra, "One way to describe what happens as their awareness grows is to say that they come to appreciate a distinction they were previously unable to make; their power of discernment increases." ([De Canson, 2024, p. 23]) Along similar lines to how Easwaran and Nielsen understand sigma algebras in the context of individual agents as the propositions an agent grasps, we can understand the elements of the sigma algebras in this context as the propositions that the theories grasp. Theories that have differently theory-laden concepts of evidence draw different distinctions and recognize different differences in the world. The sigma algebras tell us what distinctions the theories do and do not understand, can and cannot make about what the world is like. The world can present theories with the same epistemic affordance without them discerning the same evidence with it. So, we have the following explication of theory-laden evidence:

Definition 1 (Epistemic Affordances and Theory-laden Evidence)**.** For a given set of worlds Ω and theory $t = \langle \Omega, \mathcal{F}, P \rangle$, an epistemic affordance e is an arbitrary non-empty subset of Ω. The corresponding theory-laden evidence e_t is the smallest proposition E in $\mathcal{F}$ such that $e \subseteq E$.[12]

This definition formalizes the idea that two different theories may be given the same affordance by the world but, due to differences in their conceptual resources, differ in what they see. The affordances given by the world make drawing distinctions possible, but what distinctions we make out depends on us.

It is worth taking a moment at this point to address those who think the sigma algebra should be uniform over theories. They might say that if a scientific theory does not have a particular proposition (that we recognize as meaningful) in its sigma algebra then it should be extended to have it. But then the scientific theory, understood as a probability space, would have to assign a probability and conditional probabilities for the proposition. This makes odd demands on scientific practice. Why, for example, when comparing

[11] While I am inspired by the awareness growth literature when it comes to modeling conceptual resources, I am not concerned with updates that involve a change in the sigma algebra. For the purposes of this paper, the sigma algebras of scientific theories are fixed for all time, so nothing in this discussion depends on how one models awareness growth.

[12] I assume that such a smallest proposition always exists. This is so when Ω is finite. For the general case, we will require certain additional conditions.

genetic and biomechanical theories of cancer should we require the genetic theory to make predictions about biomechanical tissue organization? Why should an ecological theory that ignores a species distinction made by another theory on the basis of coloration have to make predictions about the frequency of that coloration in a population? Perhaps a scientist who grasps all of these propositions will have credences represented by a probability space in which all of these propositions are assigned probabilities and conditional probabilities. However, this point does not land for the theory-centered—rather than agent-centered—approach to Bayesianism taken in this paper. Theories, unlike agents perhaps, need not deal with all conceivable propositions, and new theories may deal with propositions not in the sigma algebra of previous theories. As Wenmackers and Romeijn put it, "Upon the introduction of a new theory, the domain of the probability function may change." (Wenmackers and Romeijn 2016, p. 1229) Now, let's return to examining the consequences for theory-ladenness of allowing the sigma algebra to vary.

This formalization can handle many different kinds of theory-ladenness of evidence—there are many reasons why a proposition might not be included in a theory's sigma algebra. A proposition might not be included simply because it is not taken to be relevant to the field of study, because the proposition is not considered meaningful by practitioners in the first place, etc. Let us see how the formalization performs on a few examples.

If theories disagree on whether some propositions are meaningful in the first place, they will disagree on what the evidence is if such propositions are used to express it. For example, the basic concepts of the humoral theory of medicine do not all map onto concepts or phenomena that modern medicine recognizes. The proposition expressing what the humoral theory grasps as "Patient has an excess of black bile" is not in the sigma algebra of the modern medical theory. In a concrete context, this proposition would contain those possible worlds that would lead a competent (by the theory's lights) practitioner of humoral medicine to conclude that the patient has an excess of black bile given a suitable epistemic affordance. For example, a competent humoral theorist might conclude an excess of black bile on the basis of what he would take to be indicators of melancholy[13] or demonic possession. These indicators may be as resistant to expression in terms accessible to the modern medical theory as the black bile concept (particularly in the case of demonic possession). Nonetheless, without taking either (or both) sides, a

[13]Note that this differs from what modern medicine calls melancholic depression.

locally neutral modeler's perspective would include such factors, whatever they may turn out to be, as differences among possible worlds that afford such a distinction to be included in the evidence. The modern medical theory will never come to draw this distinction, much less assign a probability or conditional probability to it. It is not a proposition that the theory grasps. It is not part of the theory's sigma algebra.

A closely related kind of theory-ladening of evidence comes from difference in classification schemes. We already saw an example of this in our motivating story about the ornithologists. Theories that have different classification schemes will distinguish individuals differently on the basis of the same epistemic affordances. The situation of our ornithological theories is represented in the following diagram:[14]

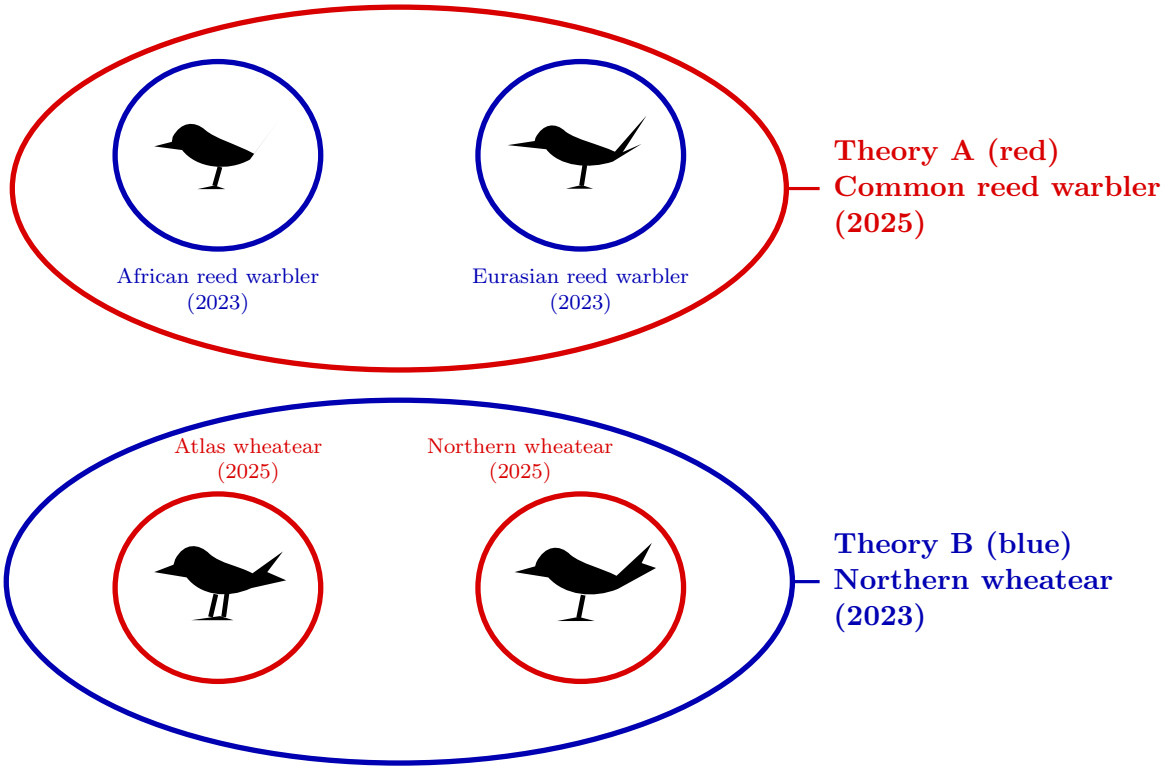


Figure 1: Distinctions grasped by Ornithological Theories

[14]tikz code for this diagram was generated with the assistance of ChatGPT-5.6. No generative artificial intelligence was used in the research or writing of any other part of this paper.

Even though the epistemic affordance makes drawing both distinctions possible, the theories disagree about which distinctions to use. Suppose individuals corresponding to the top-left and bottom-left types are presented to the ornithologists. One ornithologist's theory considers the observed individuals to be an African reed warbler and a Northern wheatear while the second ornithologist's theory considers them to be a Common reed warbler and an Atlas wheatear. Even though there is better ground for mutual understanding than in the case of humoral and modern medicine, these theories still take incompatible stances on how to distinguish or group together observed individuals. The distinctions of both theories may even be grasped at the practitioner level, but this does not mean that they must show up at the theory level. Each theory does not assign probabilities or conditional probabilities for some of the other theory's propositions, which therefore cannot show up in its sigma algebra much less act as evidence for it.[15]

Finally, another familiar kind of theory-ladenness of evidence comes from theories' determining which phenomena count as relevant and which do not. Propositions describing irrelevant phenomena are not treated by the theory, so they do not receive probabilities and therefore are excluded from the sigma algebra. Formally, the situation is the same as in propositions that are considered strictly meaningless. Informally, the reason for the exclusion is different. Similarly to differences in classification schemes, the theories do not lead their practitioners to consider the phenomena handled by the other theory meaningless, but not including them in the sigma algebra reflects that they are considered unimportant compared to the phenomena that the theory does treat. For example, two competing theories may make accurate predictions about phenomena at different scales, saying nothing about the scale at which the other theory operates. This is famously the case with general relativity and quantum mechanics. General relativity and quantum mechanics each break down at the other's scale of application. However, we don't understand this as falsifying both theories but as indicating where the theory has not been developed enough to make predictions that we take seriously. In this sense, the propositions describing phenomena at the Planck

[15]There is a close conceptual connection between *not* distinguishing certain individuals and the exchangeability property of probability measures, which de Finetti showed to be equivalent to the assumption that individuals are independently and identically distributed random samples in the real-valued case, but pursuing this interesting connection between (lack of) conceptual resources and statistical modeling assumptions is beyond our scope here.

scale for example are simply not included in the sigma algebra representing the conceptual domain of General Relativity.

3 Commensurability

The final explication we need is one for theoretical (in)commensurability. This explication draws on work in Bayesian confirmation theory, a field with roots in both philosophy (Carnap 1950) and statistics (I.J. Good 1950). In a Bayesian setting, when we want to compare two theories on the basis of some evidence we are looking to compare how well confirmed they are on the evidence. So, theories are incommensurable if there is no (consistent) concept of confirmation to deploy when making these comparisons. There is a subtle point here. Theoretical incommensurability does not just mean that there are some theories on which the comparison is undefined, i.e. some T_1 and T_2 such that neither $T_1 \leq T_2$ nor $T_2 \leq T_1$. Theoretical incommensurability means that there is no consistent concept for the $\leq$ relation to represent even where it is defined. This is what I aim to establish.

In Bayesian confirmation theory[16], a candidate quantitative explication of the concept of confirmation is a confirmation function $c(t, e)$ taking pairs of theories and pieces of evidence to the real numbers. If the theories, considering them as I do as probability spaces, have the same sigma algebra then we can treat them as propositions in a larger sigma algebra such that the probabilities conditional on them are the probabilities assigned by the respective theories. We can then also appeal to prior probabilities over the theories themselves. This is how Bayesian confirmation theory typically proceeds, but as we saw in the previous section we cannot always assume that theories have the same sigma algebra.

Since I do not assume that the theories we are comparing are defined on the same sigma algebra, there are obstacles to drawing on the existing, quantitative measures of confirmation in the literature. I will instead draw on qualitative measures of contrastive confirmation. (Carnap 1953)(Fitelson 2007)(Chandler 2013) Contrastive confirmation defines a relation so that of any two theories T_1, T_2, and evidence E, we can say whether E supports T_1 at

[16]To be clear, I am just taking for granted that a Bayesian explication of confirmation is the right one. I will not take the time to rehash arguments for and against Bayesianism. Non-Bayesians who believe in the objectivity of science are even free to read the eventual theorem as a reductio of Bayesian models of scientific inquiry.

least as well as T_2 ($T_2 \leq_e T_1$), T_2 at least as well as T_1 ($T_1 \leq_e T_2$), or supports them equally ($T_1 \sim_e T_2$). Any confirmation function $c(t, e)$ straightforwardly induces such a relation, so there can be no quantitative relation without a qualitative one. (So, not to get ahead of myself, an impossibility result for a qualitative relation is also an impossibility result for a quantitative relation.)

Fitelson (2007) introduces a variety of contrastive confirmation relations $\leq_E$ that one can get by plugging in various choices of confirmation function $c(T, E)$ in the standard setting. I will sidestep the choice of confirmation function by focusing on *local* contrastive confirmation. In local contrastive confirmation, when comparing T_1 and T_2 on evidence E, we are interested in how T_1 and T_2 compare in light of E without taking any other theories into consideration. In the standard setting, the natural way to formalize this concept is by applying one's preferred confirmation measure after conditioning on $T_1 \vee T_2$. This is the approach taken by IJ Good who uses a conditionalized quantity, $W(H : E|H \vee H')$ in his own notation, to represent "the weight of evidence in favour of H *as compared with* H', provided by E". (IJ Good 1985, emphasis added) This local contrastive confirmation concept is not widely treated in the Bayesian confirmation literature but only, I think, because there is no disagreement over it. As Fitelson himself points out, when H_1 and H_2 are logical opposites (such as when we condition on $H_1 \vee H_2$, "It is easy to see that, in such cases, we will always have $Pr(E|H_1) > Pr(E|H_2)$ iff $c(H_1, E) > c(H_2, E)$, *for any Bayesian relevance measure* c of degree of non-relational confirmation."[17] (Fitelson 2007, emphasis added) Local contrastive confirmation is not a competitor concept with contrastive confirmation. It is a particular application of the general concept, and restricting our focus to local contrastive confirmation allows the results to be agnostic to how the general concept of confirmation is formalized. Similar to the relation between quantitative and qualitative concepts of confirmation, every qualitative confirmation relation induces a local qualitative confirmation relation so we cannot have the former without the latter. (So, not to get ahead of myself again, an impossibility result for a local qualitative confirmation relation is also an impossibility result for a general qualitative confirmation relation.)

I propose to explicate theoretical commensurability with local contrastive confirmation adapted to the theory-laden context. To say that theories are

[17]Note that when Fitelson says "any" c he means those Bayesian relevance measures actually discussed in the literature and not literally any choice of function c.

commensurable is to say that we can compare how they fare in light of experience. In a Bayesian setting, this is to compare how well confirmed two theories are compared to each other in light of the epistemic affordances the world makes available. This is (or at least gives rise to as a special case) a local comparison in the sense described of not needing to take other theories into consideration when making the comparison. Adapting local contrastive confirmation to the theory-laden setting will require making a few changes. The comparison relation must be defined over theories that may not share the same sigma algebra, and it should be general enough to apply to any theories we want to compare and on the basis of any epistemic affordance. So, our explication of theoretical commensurability will take the form of a local comparison relation, defined as follows:

Definition 2 (Local Comparison Relation)**.** Let Ω be a set of possible worlds. Let T_Ω be the set of all probability spaces defined on Ω. A local comparison relation for Ω is a family of relations $(\leq_e)_{e\subseteq\Omega}$ indexed by the subsets of Ω such that each $\leq_e$ is a relation from T_Ω to itself.

We read $t_1 \leq_e t_2$ as "t_2 is at least as confirmed as t_1 on the affordance e."

Now, I will set out three conditions that I take to be implicit in how we intuitively think about comparing how well theories are confirmed. The first condition is Non-cyclicality: the induced strict comparison relation should be non-cyclic. A formalization that tells us that there are cycles of theories each strictly more confirmed compared to its predecessor is not plausible. The second condition is Locality. We saw that in the standard setting all Bayesian confirmation relations in the literature agree on the ordering of theories in terms of local comparisons, so we should continue to respect this ordering among theories that share the same sigma algebra. Our goal is to extend local contrastive confirmation to the general setting, so Locality is a minimal condition for our resulting concept to be an extension rather than a revision. The third condition, Non-punitivity, is a minimal condition for extending the concept from the standard setting of fixed sigma algebras to the theory-laden context of variable sigma algebras. Non-punitivity says that if the confirmation relation between two theories is $t_1 \leq_e t_2$, then the relation does not change if we extend t_2 to talk about things that t_1 does not grasp or vice versa, so long as this does not lead to deductive falsification.[18] The

[18]Thanks to an anonymous reviewer for pointing out the importance of this last qualification on Non-punitivity.

intuition is that t_2 should not be punished (or rewarded for that matter) compared to t_1 for sticking its neck out further by additional claims unless these claims give grounds for further comparison to t_1 (and this only so long as t_2 does not sticks its neck out and into a falsificationist guillotine.)

We thus have the following conditions for a formalization to be adequate to the intuitive comparison concept we appeal to in ordinary scientific talk.

Definition 3 (Acceptable Local Comparison Relation)**.** A local comparison relation $(\leq_e)_{e\subseteq\Omega}$ is called acceptable if it satisfies the following for all $e \subseteq \Omega$:

(I) (Non-cyclicality) For all theories $t_1, t_2, \ldots, t_n \in T_\Omega$ it is not the case that $t_1 <_e t_2 <_e \ldots <_e t_n <_e t_1$ (where $t_i <_e t_j$ means $t_i \leq_e t_j$ and $t_j \not\leq_e t_i$).

(II) (Locality) For all $t_1, t_2 \in T_\Omega$ such that t_1 and t_2 have the same sigma algebra, $P_1(e_{t_1}) \leq P_2(e_{t_2})$ if and only if $t_1 \leq_e t_2$.

(III) (Non-punitivity) For all $t = \langle \Omega, \mathcal{F}, P\rangle$, $t_1 = \langle \Omega, \mathcal{F}_1, P_1\rangle$, and $t_2 = \langle \Omega, \mathcal{F}_2, P_2\rangle$ in T_Ω, if

(i) t extends t_2 in the sense that $\mathcal{F}_2 \subseteq \mathcal{F}$ and $P \restriction \mathcal{F}_2 = P_2$

(ii) $(\mathcal{F} \setminus \mathcal{F}_2)$ is disjoint from $\mathcal{F}_1$, and

(iii) $P(e_t) > 0$

then $t_1 \leq_e t_2$ iff $t_1 \leq_e t$ (and $t_2 \leq_e t_1$ iff $t \leq_e t_1$).

These minimal conditions on any adequate formalization of the concept of local theory comparison are sufficient to show that the theory-laden context leads to the incommensurability of scientific theories by proving that there does not, in general, exist acceptable local comparison relations for arbitrary Ω. (In particular, none exist for Ω larger than a particular finite number.)

Theorem 1. *There does not exist an acceptable local comparison relation for* Ω *if* $|\Omega| \geq 2,985,984$.

For a proof, see the technical appendix.

4 Philosophical Discussion

The theorem is a formal version of the thesis of the incommensurability of scientific theories. The theorem shows that there is no consistent underlying concept of confirmation that can make sense of statements like "Theory A is better confirmed as compared with theory B given E" across all theories A, B, and epistemic affordances E, at least not while respecting the acceptability conditions. It turns out that we cannot be using one and the same concept, $\leq_e$, across all such statements unless $\leq_e$ turns out to violate one of the conditions on acceptable local comparison relations. But the conditions on acceptable local comparison relations are, I think, implicit in what we mean by such talk. A relation $\leq_e$ that violated any of the conditions would not be a faithful explication of the concept we intend.

If the intended $\leq_e$ violated Non-cyclicality, we would not blink at a claim like "Given what has happened, theory A is strictly better confirmed as compared with theory B, theory B is strictly better confirmed as compared with theory C, and theory C is strictly better confirmed as compared with theory A." If it violated Locality, then we would readily agree if someone said "In order to figure out if theory A is better confirmed as compared with theory B by what we've just seen, we must first also crunch the numbers for theories J, K, and L." Finally, if it violated Non-punitivity, we would be comfortable saying "Well, Ms. Theory-A-Proponent, you've argued a fine case against Mr. Theory-B-Proponent, but given what we've seen we currently prefer theory B. However, while Mr. Theory-B-Proponent didn't have anything to say about the particle scattering you discussed, we thought his not saying anything gave him an edge. If you're willing to retract what you said there, we'll hand the win to you!" That we are not happy with any of this talk shows that we intend a concept that satisfies all of the acceptability conditions.

But it turns out that *nothing* can answer to our intended concept in a context that allows for the theory-ladenness of evidence. The intended concept is inconsistent. Theories are incommensurable in the sense that there is no consistent concept to use when comparing them.

I expect resistance to this conclusion. One could take the conclusion as a reason to reject one of the explications of the framework. The first explication was of scientific theories as probability spaces. I argued that this explication expresses an ideal internal to scientific practice implicit in the elaboration of scientific theories into systems all of whose parts can be statistically checked and all checks of which can be coherently combined.

A great deal of work remains to clarify why, how, and how broadly this ideal can be read off of scientific practice. I have only been able to touch on this, but I trust this formalization can be simply granted here due to space constraints if nothing else. The second explication was of theory-laden evidence as the result of differences in conceptual resources represented by different sigma algebras. This explication aligns well with the epistemological research program studying awareness growth and I also argued for it by using it to model various examples of sources of the theory-ladening of evidence, so I take it to be non-problematic. Finally, the explication of theory comparison as comparisons of confirmation is very well entrenched by Bayesian Confirmation Theory, and as I am assuming a Bayesian context in this paper I don't see a way to drop this. So, the only thing we might do to modify the third explication is to revise the comparison concept by dropping one or more of the acceptability conditions, but this would concede the incommensurability of scientific theories understood in terms of the unrevised comparison concept. I will discuss this more later. A final defense tactic is to accept the applicability of the theorem but downplay its consequences. There are a few things one could say here.

First, one might say, "The theorem shows us that there is no formula to follow to determine confirmation in the theory-laden setting. This shows us that science is difficult, but we already knew that. If there's no algorithm to tell us how to interpret the evidence, we must simply muddle through as we always have." This response misses the point of the theorem. The problem is not that we do not have an algorithm for confirmation but that we do not have a consistent concept of confirmation that we could be trying to get at in the first place. Let's muddle through, by all means, but where should we be muddling to?

Second, one might say, "Subjective Bayesians accept that there is no privileged prior probability measure and that choice of prior introduces subjectivity into science. This just shows that the same is true of the sigma algebra." To this, I want to first point out that subjectivity of the sigma algebra has more bite than subjectivity of the prior. The familiar theorems showing convergence of probability measures defined over the same sigma algebra do not apply here. We also no longer have even semi-objective measures of evidence like Bayes factors. Second, as with the previous reply, I think this reply underestimates the significance of showing that there is no coherent concept of local comparison to even attempt approximating. Subjective Bayesianism in practice is tempered by the objectivity of the

likelihood function that expresses the intersubjectively-agreed support of simple hypotheses by the evidence. (Dawid 1982) One's position on the subjective–objective spectrum is a matter of whether they think statistical inference should be supplemented with subjective judgments or depend only on likelihood functions, stopping rules, etc. This arrangement does not persist in the theory-laden context. The absence of a coherent concept of local theory comparison doesn't push us closer to one end of the subjective–objective spectrum; it eliminates the spectrum by removing the concept of objectivity that holds up the other end.

Third, one might say, "Keep in view the actual scientists. If they grasp the propositions of both theories and satisfy the assumptions of a representation theorem, they will be representable as having degrees of belief on a richer sigma algebra so that if $P(e|t_1)$ is defined so too is $P(e|t_2)$. Then we do Bayesian confirmation theory as usual." To this, I should first repeat that my concern in this paper is with scientific theories and not the personal beliefs of flesh and blood scientists. However, even on its own terms I foresee problems for this approach. There will have to be some probability, conditional on the theory, for those evidence statements the theory considers meaningless. It can't assign a non-extreme probability or it would not consider the statement meaningless in the first place. If it assigns probability zero then the theory would be completely falsified by the evidence statement in favor of any theory that assigns it non-extreme probability, but this means we can't determine its truth by observation. If we could, it would be too easy to falsify a theory just by making an observation and noting that what was observed was not a meaningful observation statement according to the theory. We don't observe that statements are meaningful by looking at the world like this; what statements are meaningful to us is something we must determine prior to making observations. Finally, if the theory assigns probability one to the statement, then we have the problem just discussed with the negation. So, we are forced into requiring that evidential propositions that we can actually observe to be true are only those that all theories consider meaningful, but this pursuit of a theory-neutral observation language is a will-o'-the-wisp we have learned to avoid. Now, you might try to save the agent-centered approach by blackboxing the learning process. Say the scientists have standing credences in the probabilities of theories and other propositions. They undergo experiences and, in a way that can be modeled by Jeffrey shifts without appeal to propositions characterizing their observations, their credences in some theories raise and their credences in other theories

lower. As a descriptive claim, this might even be true (and if it isn't then so much the worse for the scientists). But this isn't a solution; it is an abstention from modeling the process that drives belief change. Blackboxing learning like this gives Kuhnian sociology of arational scientific belief change a formal model; it is not a challenge to incommensurability.

Let's now circle back to the idea of dropping one or more of the acceptability conditions. This concedes the incommensurability of theories in the sense of the inconsistency of our intuitive concept of theory comparison. The idea behind this option is to accept incommensurability of theories in this sense but proceed by replacing the inconsistent, intuitive concept with a consistent, revised concept. The situation is analogous to Bayesian Confirmation Theory, where authors have shown that different intuitive conditions on a quantitative measure of confirmation yield ordinally different measures.[19] (Crupi 2025)

Some authors in the confirmation literature take a monist position that tries to identify a unique correct quantitative measure of confirmation to adopt. Others take a pluralist position that endorses a variety of non-equivalent confirmation concepts.[20] We can similarly take a monist or pluralist attitude towards revising theory comparison for the theory-laden setting. The remainder of the paper can be read with either attitude in mind, but I would advocate pluralism. There is no one concept that can do all the work we want out of a theory comparison concept, but we can have a variety of concepts each of which does a different job.

If we drop Cyclicality, we can get a comparison concept that is diagnostically useful for theory development but which fails our expectation that theory comparison should ensure the best theories 'float to the top'. In particular, consider the local comparison relation determined by the likelihood ratio on the finest evidential proposition that *both* theories can distinguish on the basis of the affordance. That is, let $T_1 \preceq_e T_2$ iff $P_1(e_{T_1 \wedge T_2}) \leq P_2(e_{T_1 \wedge T_2})$ where $e_{T_1 \wedge T_2}$ is the least element of $\mathcal{F}_1 \cap \mathcal{F}_2$ containing e. The idea is straightforward: we compare theories on just the evidence they can both agree was learned.[21]

[19] As a side note, I do not interpret these results to analogously show the inconfirmability of scientific theories because I do not think the jointly inconsistent conditions used to characterize different confirmation measures are all implicit in the pretheoretic concept.

[20] See Crupi, Chater, and Tentori (2013, p. 195) for an overview.

[21] This is the kind of comparison done in practice by Bayesian statistics. In Bayesian statistics competing scientific theories are compared based on observables that they both make predictions about and observables that they don't both make predictions about are not used.

One problem with this relation is that the trivial theory whose sigma algebra is just Ω and $\varnothing$ will always be 'as good' as any other theory since it can only distinguish the trivial proposition which gets assigned probability one, but we do not want the trivial theory to be considered as good as any other. Another problem, as the construction in the proof of the theorem shows, is that this relation violates Cyclicality. However, this can be diagnostically useful. Violations of Cyclicality indicate that there is an opportunity to develop a theory that is superior to all of the theories in the cycle. If $T_1 \prec_e T_2 \prec_e T_3 \prec_e T_1$, then we are looking for a theory that has the advantages of T_2 on those parts of the evidence grasped by both T_1 and T_2, and similarly for each of the other theory pairs. We can't simply merge the theories since their probability assignments may disagree on propositions grasped in common, but knowing that each theory has room for improvement compared to another suggests that it is fruitful to seek some kind of unification. This is a revised concept fit for theory development, but not for ultimate evaluation to determine which theory is best.

If we drop Locality, we can get a comparison concept that models a kind of coherent updating with theory-laden evidence, but which fails to have intersubjective validity. Gallow (2014) develops a model of theory-dependent updating that, in short, takes the form of conditioning each theory on the finest evidence it can distinguish in the affordance and then shifting the probability around between theories in a way Gallow analogizes to theories competing for pots made up of the prior probabilities of theories who agree that (to use my own terminology) a particular proposition is distinguishable in the affordance. A particular prior P defined over some set of theories can determine a comparison relation such that $T_1 \preceq_e T_2$ if $P(T_1/e) \leq P(T_2/e)$ where the diagonal slash indicates updating with Gallow's theory-dependent updating method. While I do not have space to fully set out Gallow's method and walk through an example, it is easy to construct cases showing that the resulting comparison relation will be highly sensitive to the prior probabilities in theories and to which theories the prior is defined over. In particular, we can get a violation of Locality.[22] So the resulting comparison concept does not

[22]For readers who know Gallow's paper, the idea of the example is as follows. Let $\Omega = \{a, b, c\}$, the sigma algebra of T_1 and T_2 be discrete and the sigma algebra of T_3 be $\sigma(\{\{a, b\}, \{c\}\})$. We let $P_1(a) = 0.4$, $P_1(b) = 0.1$, $P_1(c) = 0.5$; $P_2(a) = 0.4$, $P_2(b) = 0.2$, $P_2(c) = 0.4$; and $P_3(\{a, b\}) = 0.6$, $P_3(c) = 0.4$. We let $e = \{a\}$ and we let P be defined by a uniform prior over the three theories, adapted to Gallow's framework. T_1 and T_2 assign the same probability to a so by Locality they should be comparatively equal. However, T_1

have the intersubjective validity that we want out of local theory comparison.

If we drop Non-punitivity, we can get a comparison concept that lets better theories float above worse ones and has intersubjective validity, but which essentially concedes to the incommensurability of scientific theories. In particular, for theories that have the same sigma algebra we can define the relation as usual according to Locality and we can leave it undefined everywhere else. This local comparison concept retains all the virtues of local comparison in the standard setting, indeed it is exactly the same when we restrict our attention to theories defined on some particular sigma algebra, but it concedes that in the theory-laden context of varying sigma algebras there is no comparing theories. The resulting comparison concept does not fit with the belief that a theory with more conceptual resources can be straightforwardly better than a theory with less conceptual resources. This concept says such comparison is impossible.

One final possibility that I have not personally investigated is using models from awareness growth to coherently extend theories to some common sigma algebra before comparing them. This is an interesting possibility, but the theorem nonetheless shows that it cannot yield a comparison relation that satisfies all of the acceptability conditions in general.

5 Conclusion

I developed a framework to investigate theory-laden evidence and theoretical incommensurability in a Bayesian setting. The framework depended on three explications: of scientific theories as probability spaces, of theory-laden evidence as the result of differences in conceptual resources represented by sigma algebras, and of theoretical comparability as local confirmation comparisons extended to the theory-laden setting of varying sigma algebras. I placed three acceptability conditions on any formalization of our intuitive concept of theory comparison and proved that the concept is inconsistent when made precise. This established the incommensurability of theories in the theory-laden setting by showing that no one concept of theory comparison can answer to our intentions across all cases of theory comparison. As a way forward, I advocated pluralism about revised comparison concepts. While we

does worse than the others on the learned proposition $\{a, b\}$ and so ends up having lower posterior probability than T_2.

cannot have one concept that does it all, we end up with (even better!) many different but respectable concepts of independent epistemological interest.

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Technical Appendix

Proof of Theorem 1

Proof. Let $|\Omega| = 12^6$ and suppose towards a contradiction that $(\leq_e)_e$ is an acceptable local comparison relation for Ω. Let us give each point in Ω a unique label in the set $\prod_{i=1}^{6}\{1,\ldots,6\} \times \{A, B\}$. So, for example, there is a point whose label is $\langle\langle 1, A\rangle, \langle 3, A\rangle, \langle 1, B\rangle, \langle 2, A\rangle, \langle 5, B\rangle, \langle 4, B\rangle\rangle$. An intuitive way to think of this is that each point represents six draws with replacement from an urn containing balls of six colors each of which is labeled either A or B.

We will define three probability measures on Ω with different sigma algebras and derive a contradiction from the properties of $(\leq_e)_{e\subseteq\Omega}$.

Let

$$\mathcal{F}_1 = \sigma\left(\prod_{i=1}^{6}\sigma(\{\{1,2\},\{3\},\{4\},\{5\},\{6\}\}) \times \sigma(\{\{A\},\{B\}\})\right)$$

This is the sigma algebra which, intuitively, draws all distinctions among the outcomes except between outcomes of colors 1 and 2. Similarly, let

$$\mathcal{F}_2 = \sigma\left(\prod_{i=1}^{6}(\sigma(\{\{1\},\{2\},\{3,4\},\{5\},\{6\}\}) \times \sigma(\{\{A\},\{B\}\}))\right)$$

which lumps outcomes of colors 3 and 4 and let

$$\mathcal{F}_3 = \sigma\left(\prod_{i=1}^{6}(\sigma(\{\{1\},\{2\},\{3\},\{4\},\{5,6\}\}) \times \sigma(\{\{A\},\{B\}\}))\right)$$

which lumps outcomes of colors 5 and 6. For later convenience, we will also define $\mathcal{F} = \wp(\Omega)$.

For $\omega \in \Omega$ let n^{ω}_{1A} denote the number of ordered pairs in the label of ω with a 1 in their first position and an A in their second position and similarly for $n^{\omega}_{1B}, n^{\omega}_{2A}, n^{\omega}_{2B}\ldots, n^{\omega}_{6A}, n^{\omega}_{6B}$.

Let P_1' be the probability measure over $\langle\Omega, \mathcal{F}\rangle$ such that

$$P_1'(\omega) \propto \left(\frac{3}{4}\right)^{n^{\omega}_{1A}}\left(\frac{1}{4}\right)^{n^{\omega}_{1B}}\left(\frac{3}{4}\right)^{n^{\omega}_{2A}}\left(\frac{1}{4}\right)^{n^{\omega}_{2B}}\left(\frac{1}{4}\right)^{n^{\omega}_{3A}}\left(\frac{3}{4}\right)^{n^{\omega}_{3B}}$$
$$\left(\frac{3}{4}\right)^{n^{\omega}_{4A}}\left(\frac{1}{4}\right)^{n^{\omega}_{4B}}\left(\frac{3}{4}\right)^{n^{\omega}_{5A}}\left(\frac{1}{4}\right)^{n^{\omega}_{5B}}\left(\frac{1}{4}\right)^{n^{\omega}_{6A}}\left(\frac{3}{4}\right)^{n^{\omega}_{1B}}$$

which uniquely specifies P_1' and let P_1 be the restriction of P_1' to $\mathcal{F}_1$. Intuitively, P_1' thinks that each draw from the urn is equally likely to be any of the six colors but that each draw has a $\frac{3}{4}$ chance of being labeled A if it is of colors 1, 2, 4, or 5 and a $\frac{1}{4}$ chance of being labeled A if it is of colors 3 or 6. Similarly, define P_2' such that it thinks each draw has a $\frac{3}{4}$ chance of being labeled A if it is of colors 1, 3, 4, or 6 and a $\frac{1}{4}$ chance of being labeled A if it is of colors 2 or 5, and P_3' such that it thinks each draw has a $\frac{3}{4}$ chance of being labeled A if it is of colors 2, 3, 5, or 6 and a $\frac{1}{4}$ chance of being labeled A if it is of colors 1 or 4. Let P_2 be the restriction of P_2' to $\mathcal{F}_2$ and P_3 be the restriction of P_3' to $\mathcal{F}_3$. Let $t_1 = \langle \Omega, \mathcal{F}_1, P_1 \rangle$, $t_2 = \langle \Omega, \mathcal{F}_2, P_2 \rangle$, and $t_3 = \langle \Omega, \mathcal{F}_3, P_3 \rangle$.

The setup is summarized in the following table, in which the probability assigned by each theory of an independent draw from the urn being labeled A given that it is of a particular color is given. It is noted on the side of the table which colors are not distinguished by each theory.

	1	2	3	4	5	6
t_1 (lumps colors 1&2)	3/4		1/4	3/4	3/4	1/4
t_2 (lumps colors 3&4)	3/4	1/4	3/4		1/4	3/4
t_3 (lumps colors 5&6)	1/4	3/4	3/4	1/4	3/4	

Table 1: Bias towards A given Color for each theory.

Note the symmetry among the theories that we get by cycling $1, 2, \ldots, 6$ by two. This symmetry ensures that the direction of the relation we find between t_1 and t_2 is the same as that between t_2 and t_3 and t_3 and t_1. Now, suppose that e is the singleton of the point labeled $\langle \langle 1, A \rangle, \langle 2, B \rangle, \langle 3, A \rangle, \langle 4, B \rangle, \langle 5, A \rangle, \langle 6, B \rangle \rangle$.

Define the restricted theories $t_1^* = \langle \Omega, \mathcal{F}_1 \cap \mathcal{F}_2, P_1^* := P_1 \restriction (\mathcal{F}_1 \cap \mathcal{F}_2) \rangle$ and $t_2^* = \langle \Omega, \mathcal{F}_1 \cap \mathcal{F}_2, P_2^* := P_2 \restriction (\mathcal{F}_1 \cap \mathcal{F}_2) \rangle$. These theories have the same sigma algebra, so by Locality we determine how they relate by looking at the Bayes factor on their shared theory-laden evidence, call it $e_{1 \wedge 2}$. This theory-laden evidence is the proposition that, to abuse the notation of our labels, $\langle \langle 1 \text{ or } 2, A \rangle, \langle 1 \text{ or } 2, B \rangle, \langle 3 \text{ or } 4, A \rangle, \langle 3 \text{ or } 4, B \rangle, \langle 5, A \rangle, \langle 6, B \rangle \rangle$. Based on the biases towards A in table 1, it is straightforward to see that $P_1^*(e_{1 \wedge 2}) \propto \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4}$ while $P_2^*(e_{1 \wedge 2}) \propto \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$ with the same normalizing constant in both. So, by Locality, $t_2^* <_e t_1^*$.

Now, when we extend t_2^* to t_2 and t_1^* to t_1, it follows from Non-punitivity that the direction of the relation does not change, so $t_2 <_e t_1$. By the symmetry of the construction, it similarly follows that $t_3 <_e t_2$ and $t_1 <_e t_3$,

violating Non-cyclicality. So we have a contradiction.

The case where $|\Omega| > 12^6$ is handled straightforwardly by lumping all of the extra worlds together with an arbitrary world of the first 12^6 worlds when defining the sigma algebras.

□